

# The Radial Address of the NFW Cusp Failure: Stacked Rotation Curve Residuals Across 149 SPARC Galaxies

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## Abstract

The Navarro–Frenk–White (NFW) density profile is known to overpredict dark matter density in the inner regions of many galaxies — the core–cusp problem. We localize this failure to a specific radius in disk-scaled coordinates. By fitting NFW, pseudo-isothermal (CIS), and Burkert dark matter profiles to 149 late-type galaxies from the SPARC database, computing fractional velocity residuals  $(V_{\text{obs}} - V_{\text{model}})/V_{\text{obs}}$  at every measured radius, and stacking across all galaxies with radii normalized by the photometric disk scale length  $r_d$ , we identify a systematic overprediction of rotation velocity by the NFW profile at  $r/r_d \approx 0.6$ . The per-galaxy median residual in the zone  $0.4 < r/r_d < 0.8$  is  $-0.099$  (Wilcoxon signed-rank  $p = 4 \times 10^{-13}$ ), with 72% of galaxies individually showing the overprediction. The CIS profile shows a weaker feature at the same location (median  $-0.021$ ,  $p = 0.001$ ). The Burkert profile shows no feature (median  $+0.006$ ,  $p = 0.98$ ), confirming the signal is cusp-specific. The NFW feature persists across Hubble types (not driven by bars), stellar mass-to-light ratios ( $Y_* = 0.3\text{--}0.8$ ), gas fractions (present in gas-poor galaxies), and disk morphology (present in smooth, regular spirals as well as irregular systems). A control zone at  $r/r_d = 2\text{--}4$  shows no systematic CIS residual ( $p = 0.45$ ). These results provide a quantitative radial benchmark for hydrodynamical simulations of cusp-to-core transformation: the NFW cusp's inner mass excess manifests most strongly at  $r/r_d = 0.6 \pm 0.1$ , a location set by the baryonic disk scale, not by halo properties alone.

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## 1. Introduction

The rotation curves of spiral galaxies provided some of the earliest evidence for dark matter (Rubin, Ford & Thonnard 1980; Bosma 1981). The density profile proposed by Navarro, Frenk & White (1996, 1997; hereafter NFW) — with its characteristic central density cusp  $\rho \propto r^{-1}$  — is a robust prediction of dissipationless cold dark matter (CDM) N-body simulations. However, observed inner rotation curves, particularly in dwarf and low-surface-brightness galaxies, frequently show solid-body rotation indicative of a flat central density core (Moore 1994; de Blok

et al. 2001; Oh et al. 2015; Katz et al. 2017). This core–cusp discrepancy is one of the most discussed small-scale challenges to  $\Lambda$ CDM (Bullock & Boylan-Kolchin 2017).

The discrepancy is well established in qualitative terms: NFW cusps predict too much inner mass, and cored profiles fit better. What has not been established is the precise radial location where the NFW failure is maximized when measured in disk-scaled coordinates across a homogeneous galaxy sample. Individual galaxy fits (de Blok & Bosma 2002; Katz et al. 2017; Li et al. 2018) report per-galaxy goodness of fit but do not stack residuals to identify universal radial features. Oman et al. (2015) characterized the diversity of inner rotation curve shapes but worked in physical units ( $V$  at 2 kpc) rather than disk-scaled coordinates. Steffen (2025) stacked SPARC accelerations in a unified radial framework but did not examine residuals from parametric profile fits.

This paper fills that gap. We fit three two-parameter dark matter profiles — NFW (cusp), CIS (core,  $r^{-2}$  outer), and Burkert (core,  $r^{-3}$  outer) — to 149 SPARC galaxies, compute fractional velocity residuals at every measured radius, normalize by the photometric disk scale length  $r_d$ , and stack. The resulting residual profile reveals where each model fails systematically, and a suite of robustness tests determines whether the signal is physical or artifactual.

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## 2. Data

We use the SPARC database (Lelli, McGaugh & Schombert 2016), which provides 175 late-type galaxies with Spitzer 3.6  $\mu$ m photometry and high-quality H I / H $\alpha$  rotation curves. The observed rotation velocity at radius  $r$  is decomposed as:

$$V_{\text{obs}}^2(r) = V_{\text{gas}}^2(r) + Y_* V_{\text{disk}}^2(r) + V_{\text{DM}}^2(r)$$

where  $V_{\text{gas}}$  and  $V_{\text{disk}}$  are computed from the observed photometry. We fix  $Y_* = 0.5 M_{\odot}/L_{\odot}$  at 3.6  $\mu$ m (McGaugh & Schombert 2014), with sensitivity tested at  $Y_* = 0.3, 0.7$ , and  $0.8$ .

Following Katz et al. (2017), we exclude galaxies with fewer than 5 rotation curve points, inclination  $i < 30^\circ$ , or quality flag  $Q = 3$  (severely disturbed kinematics), yielding  $N = 149$  galaxies with baryonic masses spanning  $10^7$  to  $10^{11.5} M_{\odot}$ . Disk scale lengths  $r_d$  are taken directly from the SPARC 3.6  $\mu$ m photometric decompositions (Lelli et al. 2016).

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## 3. Models and Fitting

### 3.1 Dark Matter Profiles

**NFW** (Navarro, Frenk & White 1996):  $\rho(r) = \rho_s / [(r/r_s)(1 + r/r_s)^2]$ . Central cusp:  $\rho \propto r^{-1}$ . Free parameters:  $\rho_s$ ,  $r_s$ .

**CIS** (pseudo-isothermal sphere; van Albada et al. 1985):  $\rho(r) = \rho_0 / [1 + (r/r_c)^2]$ . Flat core, outer slope  $r^{-2}$ . Free parameters:  $v_0$  (asymptotic velocity),  $r_c$  (core radius).

**Burkert** (Burkert 1995):  $\rho(r) = \rho_0 / [(1 + r/r_c)(1 + (r/r_c)^2)]$ . Flat core, outer slope  $r^{-3}$ . Free parameters:  $\rho_0$ ,  $r_c$ .

All three models have exactly two free parameters, ensuring a fair comparison.

### 3.2 Fitting Procedure

For each galaxy and each profile, we minimize  $\chi^2 = \sum_i [(V_{\text{obs}}(r_i) - V_{\text{model}}(r_i))^2 / \sigma_i^2]$ , where  $\sigma_i$  is the quadrature sum of the SPARC observational uncertainty and a 5 km/s systematic floor for non-circular motions (Oman et al. 2015). We use Nelder–Mead optimization with 25 random initializations per galaxy to avoid local minima.

### 3.3 Residual Computation and Stacking

For each galaxy, we compute the fractional velocity residual at every measured radius:  $\delta(r) = [V_{\text{obs}}(r) - V_{\text{model}}(r)] / V_{\text{obs}}(r)$ . Positive residuals indicate model underprediction; negative residuals indicate overprediction. Radii are normalized as  $x = r/r_d$ . We stack all data points across 149 galaxies into bins of width  $\Delta x = 0.2$  from  $x = 0$  to  $x = 10$ , computing the median residual, 16th–84th percentile envelope, and a Wilcoxon signed-rank test against zero at each bin.

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## 4. Results

### 4.1 The NFW Feature at $r/r_d \approx 0.6$

When stacked across all 149 galaxies, the NFW fractional residual shows a systematic negative dip — overprediction of rotation velocity — centered at  $r/r_d \approx 0.5$ – $0.6$  (Figure 1). The median residual at this location is  $-0.043$  with  $p = 1.5 \times 10^{-12}$  from a Wilcoxon test on all data points in the bin. The CIS profile shows a weaker feature (median  $-0.025$ ,  $p = 0.001$ ) at the same location. The Burkert profile shows no feature (median  $+0.004$ ,  $p = 0.36$ ).

## 4.2 Per-Galaxy Significance

To eliminate within-galaxy correlations, we compute one median residual per galaxy in the zone  $0.4 < r/r_d < 0.8$  and test whether the distribution of per-galaxy medians differs from zero (Figure 2).

NFW: 128 galaxies, median =  $-0.099$ , 72% individually negative, Wilcoxon  $p = 4 \times 10^{-13}$ . CIS: 128 galaxies, median =  $-0.021$ , 58% individually negative, Wilcoxon  $p = 1.4 \times 10^{-3}$ . Burkert: 128 galaxies, median =  $+0.006$ , 48% individually negative, Wilcoxon  $p = 0.98$ .

A control zone at  $r/r_d = 2-4$  shows no systematic CIS residual ( $p = 0.45$ ) and a weak NFW underprediction (median  $+0.009$ ,  $p = 0.003$ ), consistent with the expected S-shaped residual pattern of a cusp profile that overpredicts inner mass and underpredicts at intermediate radii.

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## 5. Robustness Tests

We subject the NFW feature to five independent tests.

### 5.1 Hubble Type (Bar Proxy)

The feature is present in early-type galaxies ( $T \leq 4$ , bar-prone; CIS  $p = 3 \times 10^{-7}$ ), mid-type ( $T = 5-6$ ), and late-type ( $T \geq 7$ , bar-rare; NFW  $p = 1 \times 10^{-7}$ ). It is not driven by barred galaxies.

### 5.2 Stellar Mass-to-Light Ratio

The feature persists at  $Y_* = 0.3$  (CIS median  $-0.020$ ,  $p = 6 \times 10^{-4}$ ; NFW median  $-0.043$ ,  $p = 1 \times 10^{-9}$ ),  $Y_* = 0.5$ ,  $Y_* = 0.7$ , and  $Y_* = 0.8$ . The sign never flips. Amplitude increases monotonically with  $Y_*$ , consistent with the disk contribution amplifying the baryon–DM coupling at this radius.

### 5.3 Gas Fraction

The feature is present in gas-poor galaxies ( $f_{\text{gas}} < 0.3$ ; CIS  $p = 0.003$ , NFW  $p = 0.006$ ) and gas-rich galaxies. It is not driven by pressure support in gas-dominated dwarfs.

### 5.4 Disk Morphology

Using the deviation of  $R_{\text{eff}}/R_{\text{disk}}$  from the pure exponential value (1.678) as a smoothness proxy, we split galaxies into smooth (close to exponential) and lumpy (deviating). The NFW feature is present in smooth disks ( $p = 2 \times 10^{-8}$ ) and is not stronger in lumpy disks (Mann–Whitney  $p = 0.86$ ). Using Hubble type, the feature is present in regular spirals ( $T = 3-6$ ; NFW  $p = 4 \times 10^{-5}$ , CIS  $p = 2 \times 10^{-4}$ ). It is not a photometric artifact of irregular morphology.

## 5.5 Burkert Profile Control

The Burkert profile, which has a flat core like CIS but a steeper  $r^{-3}$  outer fall-off, shows no feature at  $r/r_d = 0.6$  (per-galaxy Wilcoxon  $p = 0.98$ ). This confirms the feature is cusp-specific — it arises from the NFW  $r^{-1}$  inner density slope, not from any property shared by all dark matter profiles.

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## 6. Discussion

### 6.1 Relation to the Core–Cusp Problem

The core–cusp discrepancy has been established through individual galaxy fits (de Blok et al. 2001; de Blok & Bosma 2002; Oh et al. 2015) and population-level analyses (Oman et al. 2015; Katz et al. 2017). The present work does not discover the core–cusp problem. Rather, it provides a spatial localization: the NFW cusp's excess inner mass manifests most strongly at  $r/r_d = 0.6 \pm 0.1$  in stacked residuals, with per-galaxy significance at  $p = 4 \times 10^{-13}$ . This localization in disk-scaled coordinates is a new empirical result.

The location  $r/r_d \approx 0.6$  is deep within the baryonic disk, well before the disk velocity peak at  $r \approx 2.2 r_d$ . At this radius, the baryonic contribution is rising and the NFW cusp contributes additional mass that is absent in cored profiles. The overprediction is the direct kinematic signature of the cusp's inner mass excess, viewed at the radius where it is most tightly constrained by the data.

### 6.2 Implications for Simulations

Hydrodynamical simulations that model cusp-to-core transformation via baryonic feedback (Pontzen & Governato 2012; Di Cintio et al. 2014; Chan et al. 2015) can use the stacked residual profile as a benchmark. A successful simulation should produce rotation curves whose NFW-fit residuals at  $r/r_d \approx 0.6$  are consistent with zero — that is, the feedback should eliminate the cusp's excess inner mass specifically at this disk-scaled radius. The per-galaxy detection rate (72%) and median amplitude ( $-0.099$  fractional residual) provide quantitative targets.

### 6.3 Why $r/r_d$ ?

The choice to normalize by disk scale length rather than a halo property ( $r_s$ ,  $r_{200}$ ) is deliberate. The cusp failure is expressed in disk-scaled coordinates because the baryonic disk provides the constraint: at  $r/r_d \approx 0.6$ , there are sufficient data points with small enough error bars to reveal the mismatch. In halo-scaled coordinates, the failure location would vary across galaxies because the halo scale radius  $r_s$  is poorly constrained and degenerate with  $\rho_s$ . The disk provides a stable reference frame.

## 6.4 Caveats

The analysis uses a fixed stellar mass-to-light ratio ( $Y_* = 0.5$ ) and Nelder–Mead optimization. While the feature persists across  $Y_* = 0.3$ – $0.8$ , future work should verify with per-galaxy Bayesian fitting (cf. Li et al. 2018) and MCMC methods. Replication on independent datasets such as THINGS (Walter et al. 2008) and LITTLE THINGS (Hunter et al. 2012) is encouraged. The CIS detection is marginal ( $p = 0.001$ , 58% of galaxies) compared to the NFW detection ( $p = 10^{-13}$ , 72%) and should be treated as secondary.

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## 7. Conclusions

1. The NFW cusp failure in SPARC rotation curves peaks at  $r/r_d = 0.6 \pm 0.1$  when residuals are stacked in disk-scaled coordinates. Per-galaxy median residual:  $-0.099$ , Wilcoxon  $p = 4 \times 10^{-13}$ , 72% of 128 galaxies individually affected.
2. The feature is cusp-specific: CIS shows a marginal signal ( $p = 0.001$ ); Burkert shows none ( $p = 0.98$ ).
3. The feature survives five robustness tests: it is not driven by bars, mass-to-light ratio assumptions, gas fraction, or irregular disk morphology.
4. The stacked residual profile and per-galaxy statistics provide a quantitative benchmark for hydrodynamical simulations modeling cusp-to-core transformation.

Data available via the SPARC public archive: <http://astroweb.cwru.edu/SPARC/>. Analysis code accompanying this paper is available in the supplementary files.

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